Shopping list and cut schedule

Four separate plywood boxes | 93 wide x 16 high x 20 deep before trim and joined white top

BUY FOR THE FOUR CABINET BOXES

3 sheets of nominal 3/4" cabinet plywood cover the original boxes, shelves, surround and one door blank.
Reserve additional stock for the joined white top and remaining three doors; finalize material and thickness first.
Do not buy the former 10 ft butcher-block blank. Finished top target: 112" x about 20 1/4", field-fit.
Trim: two straight 8 ft poplar strips for long rails, plus matching stock for five short strips.

CUT FROM SHEETS 1 AND 2

ID	PART	QTY	FINISHED SIZE OR BLANK
A/B	Xbox and Denon tops and bottoms	4	26 x 20
C	Switch 2 top and bottom	2	20 x 20
D	Storage top and bottom	2	21 x 20
S	Sides, two per box	8	H x 20 | calculate H below
SH	Storage shelf and optional Switch shelf	2	20 x 20 blanks | fit after assembly

MEASURE THE PLYWOOD BEFORE CUTTING THE SIDES

H = 16 minus the actual thickness of that box's top and bottom combined.
True 3/4" stock gives H = 14 1/2". True 23/32" stock gives H = 14 9/16".
These cuts assume square butt joints, with the sides fitted between full-width tops and bottoms.

Match thicknesses across the boxes. If you change to joinery that recesses either panel into another, recalculate the affected cut dimensions first.

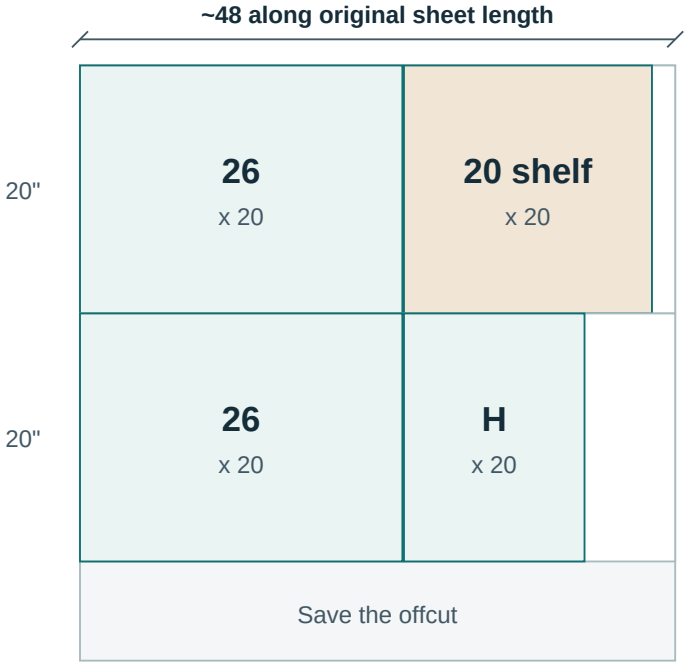
Build sequence: boxes p. 6 | mounting strips p. 7 | install p. 8 | finish and alcove pp. 9-11 | Sheet 3 p. 12

Plywood breakdown

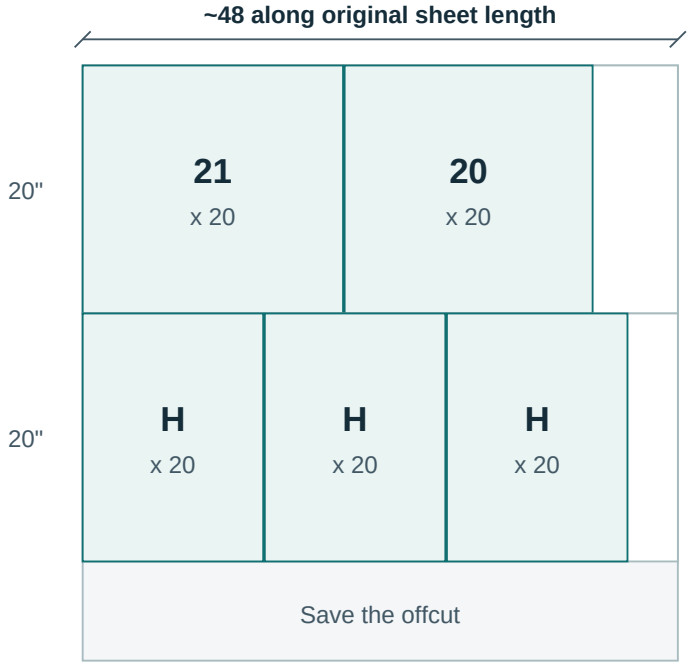
Repeat the same layout on Sheet 1 and Sheet 2. Sheet 3 stays whole.

For transport, ask for one rough crosscut near the midpoint of each 96-inch sheet.
Keep both halves at least 47 inches long along the original 96-inch direction.

HALF A / ONE PER SHEET



HALF B / ONE PER SHEET

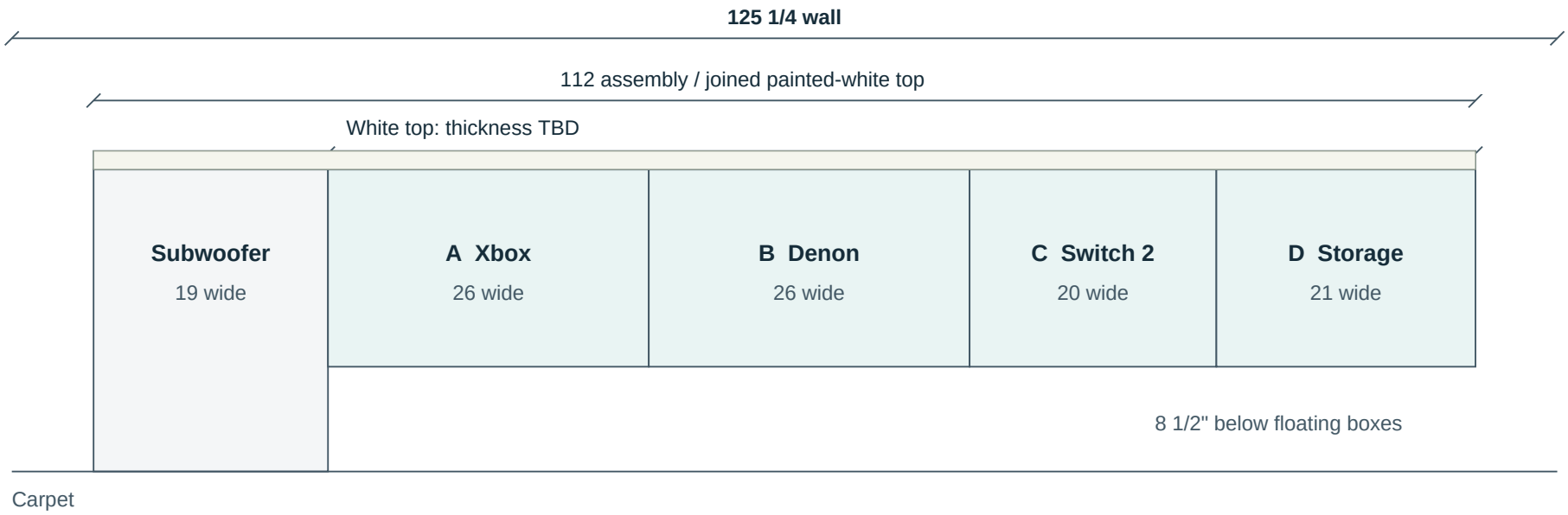


1. Square a reference end and edge. Rip two 20-inch strips from each half along the original sheet length.
2. Crosscut the parts in the order shown. H is the measured side height from page 1.
3. Measure each part from the new cut edge and put the saw kerf on the waste side. Label the parts by bay.

The longest strip needs 46 inches of parts. The 47-inch minimum allows 1/8-inch kerfs and a light squaring cut.

Front elevation

Centered on the 125 1/4-inch wall | Four boxes and one separate subwoofer surround



WALL POSITIONS FROM THE LEFT EDGE

SUBWOOFER	A / XBOX	B / DENON	C / SWITCH 2	D / STORAGE
6 5/8 to 25 5/8	25 5/8 to 51 5/8	51 5/8 to 77 5/8	77 5/8 to 97 5/8	97 5/8 to 118 5/8

REFERENCE ELEVATIONS ABOVE CARPET

Box bottom 8 1/2". Box top 24 1/2". Finished height = 24 1/2" + selected top thickness.

Draft outlet group is 56 1/2 to 63 1/4 from the left and 9 3/4 to 14 1/2 above carpet.

Verify the wall, studs and outlets on site before using these positions for mounting or openings.

Current direction: four inset doors, one per floating bay; subwoofer stays open. See pages 13-14.

Parts to fit after assembly

Measure the installed boxes before cutting finish parts, hardware openings or the wall support.

01 Poplar face trim

Use 1/4" thickness for the slim front we discussed. Fit two continuous horizontal rails across the 93" run.
Cut three seam covers and two end strips to fit between the rails. Seam covers are about 1 1/2 wide.
Match each strip to the plywood edges so the trim sits flush with the bay openings. Size rails to cover the edges.

02 Shelves and four inset doors

Fit the shelf blanks to the assembled openings. Target about 18" depth where the hardware and cables permit.
Fit one inset door per bay (four total). Subtract 1/4" from each measured opening for 1/8" reveals.
Verify that reveal with the chosen hinges. Test the 35 mm hinge cups and plate setback on scrap with the trim.
For a knob or pull and self-closing hinges, set the adjustable stop after aligning the door flush with the trim.

03 Joined painted-white top

Join shorter panels into one finished white top spanning all 112". Place the seam over solid support.
Target 112" x about 20 1/4"; material and thickness TBD. Finished height = 24 1/2" + top thickness.
Set the alcove support flush with the plywood tops so it supports the cap independently of the subwoofer.
Retain all box tops. Stabilize the seam mechanically before filling, sanding, priming and painting. See p. 14.

04 Backs, surround and wall support

Keep the Denon rear open for outlets, plugs and airflow. Size any removable liners and storage back after test fit.
Keep the subwoofer floor-supported. Measure it with cords attached before cutting its separate surround.
See pages 5-8 for the ledger, box assembly, mounting strips and direct fastening sequence.

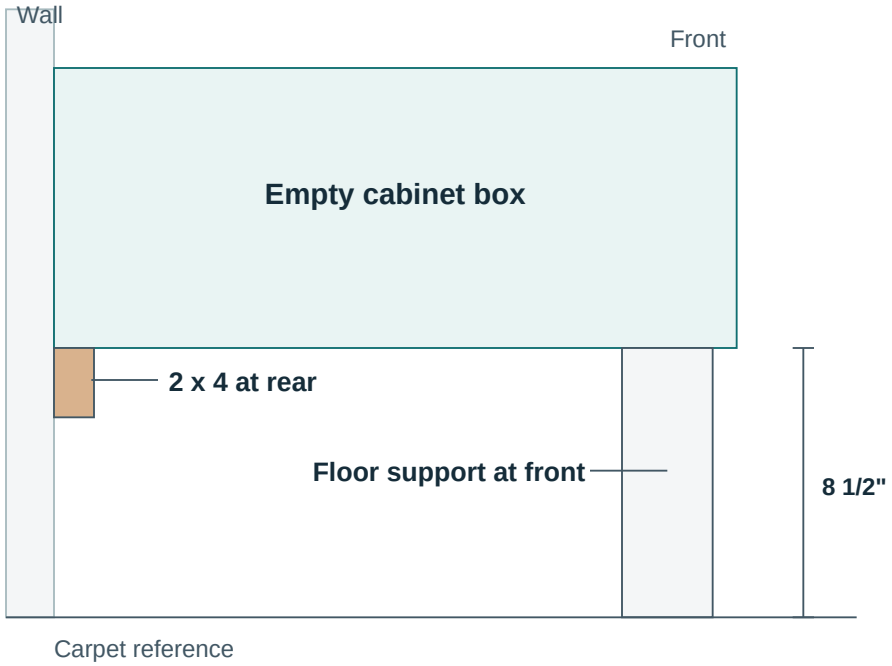
Temporary installation ledger

Direct mounting to wood studs and in-wall blocking | Install the boxes one at a time

CUT ONE STRAIGHT NOMINAL 2 x 4 TO 93 INCHES

Buy one 8 ft board. Place its wide face against the wall so it projects 1 1/2" under the rear of each box.
Set its top edge at the 8 1/2" cabinet-bottom line. The planned span is 25 5/8" to 118 5/8" from the left wall.
Transfer one level line from the carpet reference used on page 3. Check floor clearance across the run.

SIDE VIEW / SCHEMATIC



SET UP AND FASTEN

1. Mark verified wood studs at ledger height and check for wiring below the Denon outlets. Secure the ledger into studs with suitable wood fasteners.
2. Rest the rear edge of one empty box on the ledger. Use a stable, floor-supported platform under the front at the same height. Have a helper steady the box.
3. Level and shim the box. Drive permanent fasteners through its upper and lower rear strips into verified studs or secured in-wall blocking. Repeat for each box.
4. Align and join the boxes. Once permanently secured, remove the ledger and platform, patch the screw holes, then fit the alcove support and cap before loading.

The ledger is temporary. Confirm permanent blocking, strip joinery and wall fasteners for the loaded cabinet.
Keep the Denon outlet plates and plugs accessible when locating its lower attachment.

Reference: Cardell installation guide, temporary ledger and front brace

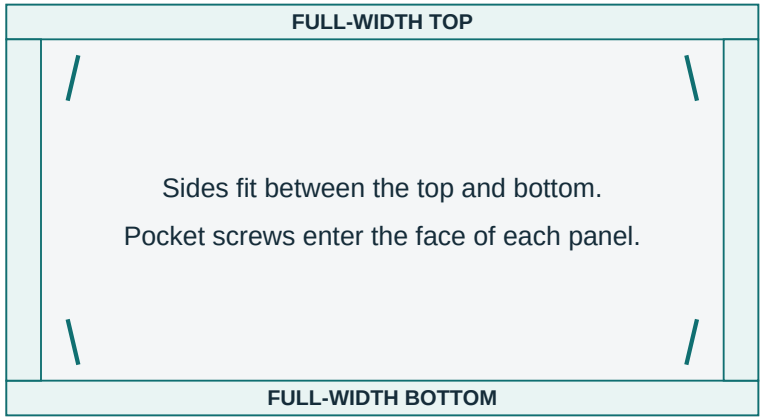
Assemble the four plywood boxes

Build A, B, C and D separately. All boxes are 16 high x 20 deep before finish trim.

USE KREG POCKET HOLES FOR THE PLYWOOD BOX JOINTS

Use coarse-thread screws for plywood. A 1 1/4" Kreg screw is the starting size for true 3/4" plywood.
Set the jig and drill stop for your measured stock. Test on two offcuts before drilling the finished parts.

FRONT VIEW OF ONE BOX



Pocket directions shown. The drawing omits the rear strips.

01 Mark and drill

Label each panel by box, front edge and inside face.
Drill pockets on the inside face near both ends of each side.
Distribute them along the 20-inch depth and keep the rear strip and cleat hardware clear of the pocket locations.

02 Glue, clamp and drive

Dry-fit the four panels first. Apply wood glue to mating edges, clamp on a flat bench, and align the front edges.
Drive the pocket screws with a drill clutch set low enough to seat the heads cleanly without stripping the plywood.

03 Check for square

Compare both diagonals and adjust until they match.
Check for twist on the bench and confirm the box is 16 high.
Dry-fit the rear strips while the box is square.

CHECK EACH BOX BEFORE MOVING ON

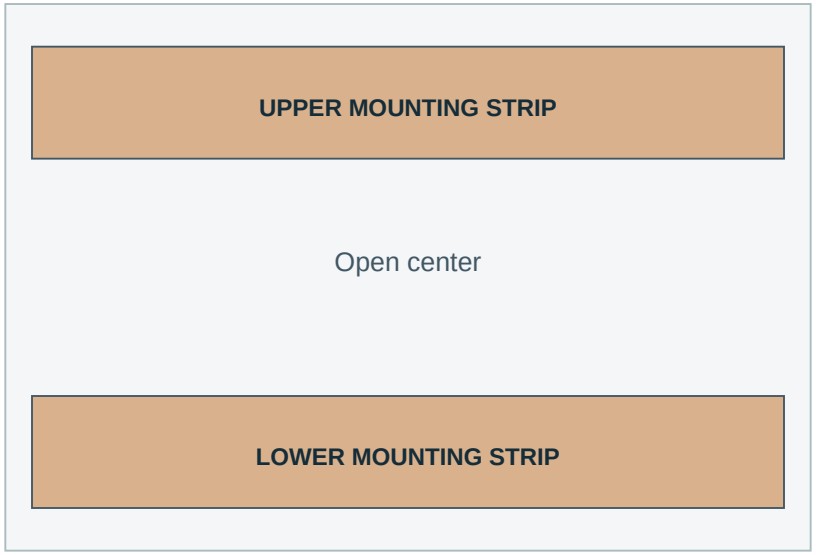
Confirm the outside width and depth, flush front edges, and square corners. Fit strips to each actual opening.

Kreg pocket screw selection and jig setup

Build in the mounting strips

One upper and one lower strip in each box | Standard detail assumes a clear rear opening

REAR VIEW / ONE BOX



Wide faces face the wall. Fit the rear faces flush with the box.

CUT AFTER MEASURING THE BOXES

Fit both strips to the assembled box

Use straight, dry solid stock. Nominal 2 x 4 is the layout allowance shown here (1 1/2 x 3 1/2 actual). Set the upper strip against the underside of the top and the lower strip on the top face of the bottom.

Tie the strips into the box

Fit solid-wood cleats inside at the strip ends. Each cleat bears on the inside face of the side and on the strip. Glue bare mating faces and mechanically fasten into both members. Tie the upper strip to the top and the lower strip to the bottom with screws through the outer plywood face into the solid strip. Verify the joint.

Lay out fasteners before drilling

Keep cleats, pocket screws and wall screw paths clear of one another. Pocket screws may help with assembly. Verify the strip joints and fasteners for the hanging load.

A / 2 at ~24 1/2	B / 2 at ~24 1/2	C / 2 at ~18 1/2	D / 2 at ~19 1/2

Strip length = outside box width minus both actual side thicknesses. Lengths above assume true 3/4 sides.
Allow two straight 8 ft 2 x 4s for the eight strips if this stock is approved, plus separate stock for the cleats.
Dry-fit the strips and all gear before assembly. Adapt the Denon lower strip to the actual outlet opening.

The temporary 93-inch 2 x 4 on page 5 is a separate board.

AWI reference for structural cabinet anchor strips

Mount and join the cabinet boxes

Direct fastening through the rear strips into wood studs and secured in-wall blocking

LAY OUT PERMANENT FASTENERS BEFORE PATCHING THE WALL

Use the wall positions on page 3 and the 8 1/2" cabinet-bottom line. Transfer the actual rear strip locations. Provide spaced upper and lower fastening points for each box. Add blocking where those points miss studs. Verify blocking attachment, strip joints, and wall fasteners for the combined box, cap and equipment load.

01 Open and document the wall

Check for wiring and shut off the affected circuit before opening the wall near the outlet group. Secure the blocking to the framing. Record the stud centers and blocking limits from the left wall and the same height reference, photograph them with a tape in view, then patch the drywall.

02 Set each box on temporary supports

Use the rear ledger and front platform on page 5. Set an empty box in place with a helper. Align it to the elevation, check level and plumb, and shim at the anchorage points as needed.

03 Fasten and align

Fasten through the upper and lower strips into verified studs or blocking with specified cabinet anchorage screws. Check the next box against the first. Clamp adjacent sides so the faces sit flush and join them with cabinet connector screws sized for the two side panels. Verify that the screw tips stay inside the paired panels.

04 Remove temporary support

Confirm the permanent attachments in every box before removing the temporary ledger and platform. Patch the ledger screw holes. Keep all equipment and the slab off the boxes until the supports are ready.

AWI wall cabinet anchorage reference

ESFI guidance for cutting and drilling near wiring

Fit shelves, trim and four inset doors

Finish the visible parts after the boxes are aligned and the actual openings are known

01 Dry-fit equipment and shelving

Place the receiver and consoles in their bays with cables attached. Check rear clearance and ventilation. Trim the two shelf blanks to the finished openings. Allow clearance for shelf hardware and door swing. Use about 18" shelf depth where the hardware, airflow and cables permit. Fit storage backs or removable liners only after the wall attachments are checked. Keep Denon plugs and fasteners accessible.

02 Fit the poplar face trim

Cut two continuous horizontal rails from 8 ft stock. Fit the three vertical seam covers and two end strips between them. The trim is 1/4" thick. Size each piece to cover the plywood edges and preserve bay openings. Dry-fit the rails and stiles with the boxes clamped flush. Glue to clean wood, clamp or pin, then fill and sand.

03 Build and hang four inset doors

Fit four inset doors to the installed trim openings; Shaker styling is shown in the current reference. Material, thickness and hardware remain to be selected before door cutting and hinge drilling. For 1/8" reveals, subtract 1/4" from each opening dimension; confirm against hinge specifications. Test on scrap. Provide room-air intake and exhaust for enclosed equipment; see page 14.

04 Prep and paint

Fill plywood edges and visible fastener holes, sand, prime, then apply warm white cabinet paint. Paint and let parts cure before fitting lights, door hardware and electronics. Keep glue surfaces bare until the trim has been attached. Protect the painted face while fitting the top.

ORDER OF FIT

Align boxes, fit shelves and trim, measure all four doors, then finish. Keep doors removable for adjustment.

Blum inset hinge types and specifications

Support the alcove and fit the cap

One joined painted-white top | Current finish and top details on pages 13-14

01 Build the separate alcove support

Lay out the 19" left zone from page 3. Test-fit the floor-standing subwoofer with plugs and cables attached.
Use the field-fit surround detail on page 11. Measure the actual unit and allow cable and vent clearance.
Build the surround so its load-bearing top is in the same plane as the cabinet tops, about 24 1/2" above carpet.
Verify the surround carries its share of slab weight independently. Leave clearance around the subwoofer.

02 Join and finish the white top

Field-fit the 112" length and approximately 20 1/4" depth after the boxes and surround are aligned.
Locate the panel seam over a supported cabinet area. Stabilize the joint with substrate-appropriate joinery and backing where needed, then fill, sand, prime and paint. Filler is cosmetic, not the joint connection.
Retain the plywood box tops. Finalize stock, thickness, joinery and fastening before cutting; see page 14.

03 Fit lighting and route cables

Retain the warm underside lighting; interior lights are optional behind opaque doors. Test wiring first.
Provide a low room-air intake and a separate exhaust back to the room; inter-bay holes alone do not cool.
Keep drivers, fans, controllers and cables accessible and clear of hinges, fasteners and hot equipment.

04 Load and check

Check that the boxes are secure, the face seams are flush, and the door moves freely.
Set the equipment in place, verify plug access and airflow, and test the lighting and speakers.
Make the final fit check with the subwoofer standing on the carpet, clear of the surround.

FIELD-FIT PARTS

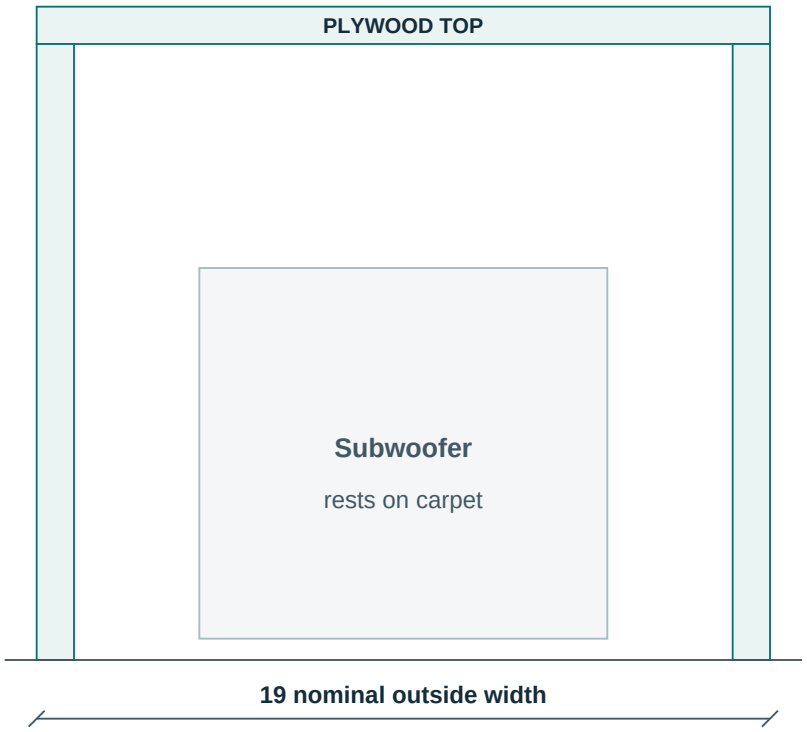
The alcove support, Denon outlet opening, and permanent fastener schedule depend on the actual wall and gear.

Current joined-top and ventilation details: page 14 (replaces butcher-block guidance).

Assemble the subwoofer surround

Field-fit detail | Separate floor-supported surround with an open front and bottom

FRONT VIEW / SCHEMATIC



Cap and rear bracing omitted for clarity.

01 Measure the actual opening

Start with the nominal 19" wide x 20" deep zone. Check the subwoofer with its cables connected and the actual front trim in place. Measure the floor at both support points before cutting the sides.

02 Cut and assemble

Fit a full-width plywood top and two sides from page 12. Side height = installed cabinet top elevation minus the top panel thickness and any firm leveling pad under that side. Use the pocket-hole procedure on page 6 for the top joints.

03 Brace and level the surround

Fit a rear stretcher between the sides near the top and secure it into both sides to keep the surround square. Keep the rear cord and vent area clear. Level on stable bearing points and verify the bracing and wall restraint for the cap load.

04 Check fit before setting the slab

Set a straightedge across the cabinet tops and surround. Adjust until both supports share a plane and the subwoofer has clearance from every surround panel.

FIT TO THE INSTALLED CABINETS

Use the measured cabinet-top height for the side cuts. The 24 1/2-inch elevation is the drawing reference.

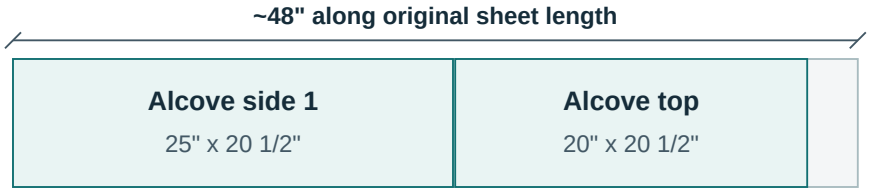
Cut the third plywood sheet

Original Sheet 3 layout: surround and ONE door blank only. Extra door/top stock required.

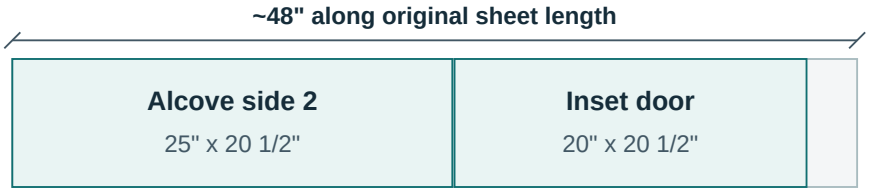
MAKE ROUGH BLANKS, THEN TRIM TO THE INSTALLED OPENINGS

Option for transport: crosscut the sheet near the middle of its 96" length. Keep both halves at least 47" long. Square one edge and end on each half, then rip one 20 1/2" strip along the original sheet length.

HALF A



HALF B



Crosscut both strips at 25" and 20". Each uses 45 1/4" with two 1/8" kerfs, before end squaring.

From a wide remainder, cut one 20" x 5" rough rear stretcher blank. Save the other offcuts.

TRIM TO FIT

Alcove top	Target 19" wide x 20" deep. Match the installed depth and the width of the surround.
Alcove sides	Target 20" deep. Height = local floor-to-cabinet-top height minus top thickness and leveling pad.
Rear stretcher	Fit between the sides. At 19" outside and true 3/4" sides, the length is 17 1/2".
Inset door	Measure after face trim. For a 1/8" reveal on each edge, subtract 1/4" from width and height.

With a 24 1/2" cabinet-top height, true 3/4" plywood and zero pad thickness, the alcove sides would be 23 3/4" tall.

Measure both sides separately. Fit the stretcher height and placement to the bracing and cable clearance on page 11.

Reserve extra stock for three more doors and the joined top; this layout is not the full revised shopping list.

Current design: four doors and a white top



Four floating bays: Xbox 26", Denon 26", Switch 20", storage 21". Separate 19" open subwoofer alcove.

The image shows the selected finish and door arrangement. Top thickness is illustrative; dimensions govern cuts.

Rev F replaces the butcher-block purchase and one-door finish plan. Box panel dimensions are unchanged.

Joined white top: decisions and assembly

01 Confirm the substrate and finished height

Target one finished 112" x approximately 20 1/4" top, made from shorter joined panels.
Material and thickness are not selected. Paint-grade plywood or MDF need different edge and joint treatment.
Keep all existing box tops and the surround top. Finished elevation = 24 1/2" + chosen top thickness.
The previous 26" height applies only to a 1 1/2" top. Reserve additional top and door stock before buying.

02 Support and stabilize the seam before filling

Dry-fit after mounting, leveling and joining the boxes. Place the seam over solid support, away from a box joint.
Use joinery suitable for the substrate and backing where needed; keep both panel faces coplanar.
Biscuits or splines can help alignment; filler alone cannot connect the panels or prevent movement cracks.
Check fastener depth and access from below. Do not cut mounting strips, cleats or box joints for seam hardware.

03 Fill, sand, prime and paint

Follow the selected adhesive cure time. Fill the stable seam with a compatible paint-grade filler and sand flush.
Seal exposed edges, prime and sand as the coating system requires, then paint the top uniformly white.
Finish exposed faces and underside as appropriate for the substrate. Let coatings cure before placing speakers.
A filled seam can telegraph later if the panels move; plan the joint and support before committing to a finish.

04 Four doors and enclosed-equipment cooling

Current visual direction: one inset Shaker-style white door on each floating bay; subwoofer stays open.
Fit doors after face trim. Three additional doors and top stock are not included in the original three-sheet layout.
Select door material, hinges and opening method; do not drill from the rendering. Verify hardware on scrap.
Provide low intake air from the room and exhaust back into the room, with fans if required for the heat load.
Inter-bay openings only connect the airflow path. Confirm equipment clearances and test temperatures closed.
Vent routes, fan sizes and structural mounting details remain to be finalized before drilling or loading.